PROPER-OVER-BASE CLOSED SUBSETS PULL BACK ALONG BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1130-1140, immediately following the statement at 1115-1128.

Editorial note. The fiber-product identity in the middle is the span's own computation; properness of Z -> Y over Y comes from the hypothesis via the defining lemma.

Allowed standard prerequisites (editorial).

- Locally finite type is preserved by base change. Morphisms of Spaces, Lemma spaces-morphisms-lemma-base-change-finite-type, cited at proof lines 1131-1133. Makes the conclusion meaningful for f' (the definition requires locally finite type).
- Properness is preserved by base change. Morphisms of Spaces, Lemma spaces-morphisms-lemma-base-change-proper, cited at proof lines 1138-1140. Z' -> Y' is proper as the base change of Z -> Y.

Source reference. Original reference 03XH: focused-seeds/geometry/S9-perf-base-change-closed-proper-over-base/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04WP: focused-seeds/geometry/S9-perf-base-change-closed-proper-over-base/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Consider a cartesian diagram of algebraic spaces over S*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

with f locally of finite type. If T is a closed subset of $|X|$ proper over Y, then $|g'|^{-1}(T)$ is a closed subset of $|X'|$ proper over Y'.

Proof. Observe that the statement makes sense as f' is locally of finite type by Morphisms of Spaces, Lemma 03XH. Let $Z \subset X$ be the reduced induced closed subspace structure on T. Denote $Z' = (g')^{-1}(Z)$ the scheme theoretic inverse

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image. Then $Z' = X' \times_X Z = (Y' \times_Y X) \times_X Z = Y' \times_Y Z$ is proper over Y' as a base change of Z over Y (Morphisms of Spaces, Lemma 04WP). On the other hand, we have $T' = |Z'|$. Hence the lemma holds. $\square$